Paleomagnetism of the Callander Complex and the Cambrian apparent polar wander path for North America

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The 7 km^2 circular Callander alkaline complex was emplaced into anorthositic and granitic gneisses of the Grenville Province in the Canadian Shield about 575 ± 5 Ma ago at the start of the Cambrian. The complex has not been subsequently metamorphosed or tilted. Detailed alternating-field and thermal step demagnetization of 252 specimens from 29 sites led to the identification of a characteristic A magnetization component with a direction of $D = 82.2°$, $I = 82.7°$ ($\alpha_{95} = 3.1°$, $k = 83$, $N =$ 26 sites) in 5 sites of mesocratic to leucocratic syenite from the core of the complex, in 5 sites of fenitized host rock from its aureole, and in 16 sites of lamprophyre from radiating dikes. Isothermal remanent-magnetization tests show that the A component is retained by both magnetite and hematite in a complete spectrum of domain sizes. A reversals test suggests and a contact test shows the A component to be primary. Its pole position at 46.3°S, 121.4°E (dp = 5.9°, dm = 6.1°) does not fall on published but poorly defined Cambrian apparent polar wander paths, leading to speculation on an alternative Cambrian path.

La mise en place du complexe alcalin de Callander, ayant une aire circulaire de 7 km^2, dans les gneiss anorthositiques et granitiques de la province de Grenville du Bouclier canadien, date du début du Cambrien, soit avec un âge d'environ 575 ± 5 Ma. Le complexe n'a pas été métamorphisé, ni dérangé par un basculement tectonique. La désaimantation par champ alternatif et par chauffage de 252 spécimens provenant de 29 sites fournit une composante d'aimantation A avec une direction de $D = 82{,}2°$, $I =$ 82,7° ($\alpha_{95} = 3{,}1°$, $k = 83$, $N = 26$ sites) pour 5 sites de syénite mésocratique à leucocratique localisées au coeur du complexe, pour 5 sites de la roche encaissante féntisée qui enrobe la syénite, et pour 16 sites de lamprohyre formant les dykes radiaires. Les tests d'aimantation rémanente isothermique révèlent que la composante A est portée par la magnétite et également par l'hématite, indépendamment de la grosseur des domaines et sans discontinuité. Les tests d'inversion et de contact démontrent que la composante A est primaire. La position de son pôle à 46,3°S, 121,4°E (dp = 5,9°, dm = 6,1°) ne tombe pas sur les courbes de dérive apparente des pôles du Cambrien qui ont été publiées mais qui sont imprécises, ce qui laisse entrevoir la possibilité d'une dérive différente au Cambrien.

[Traduit par la rédaction]

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Introduction

The Callander alkaline carbonatite complex was emplaced into the Precambrian Shield of northern Ontario at the start of Cambrian time (Fig. 1). Except for minor uplift, it has not been tectonically disturbed or metamorphosed, making it a good target to provide a basal Cambrian pole position. Further, there is little agreement on the Cambrian segment of the apparent polar wander path (APWP) for North America because the poles are widely scattered (Watts *et al.* 1980; Van der Voo 1990). Most of them come from sedimentary units, and many of these units appear to have a secondary remagnetization. Nevertheless, a reliable Cambrian APWP would be valuable to help resolve continental reconstruction problems such as the location of the Avalon terrane relative to the craton in the Cambrian (Schenk 1978).

Geology

The Callander Complex is a circular pluton of about 3 km in diameter that underlies Callander Bay at the east end of Lake Nipissing (Fig. 2). It intrudes high-rank anorthositic and granitic gneisses of the upper Middle Proterozoic Grenville Province in the Canadian Shield (Lumbers 1971). It was emplaced with four similar alkaline complexes along the Nipissing branch of the Ottawa – St. Lawrence graben system during a period of crustal extension and rifting (Kumarapeli and Saull 1966) (Fig. 1).

Drilling indicates that the core of the complex contains layered cone sheets of nepheline syenite and carbonatite. The outer shell of leucocratic and mesocratic nepheline syenite crops out along the east side of Callander Bay (Ferguson and Currie 1972) (Fig. 2). Numerous lamprophyre (<10 m width), carbonatite (<3 m), and potassium trachyte (<5 cm) and a few nepheline syenite dikes radiate from the core with various cross-cutting relationships into the surrounding fenetized host rock. The fenite aureole has a metasomatized inner zone with hematite in which gneissosity and igneous textures are destroyed, a middle zone with exaggerated gneissosity from soda pyroxene and carbonatite veinlets, and an outer zone with minor breccia veinlets and hematized cracks (Currie and Ferguson 1971, 1972). The lamprophyre dikes contain phenocrysts of altered olivine, clinopyroxene, biotite, and accessory magnetite

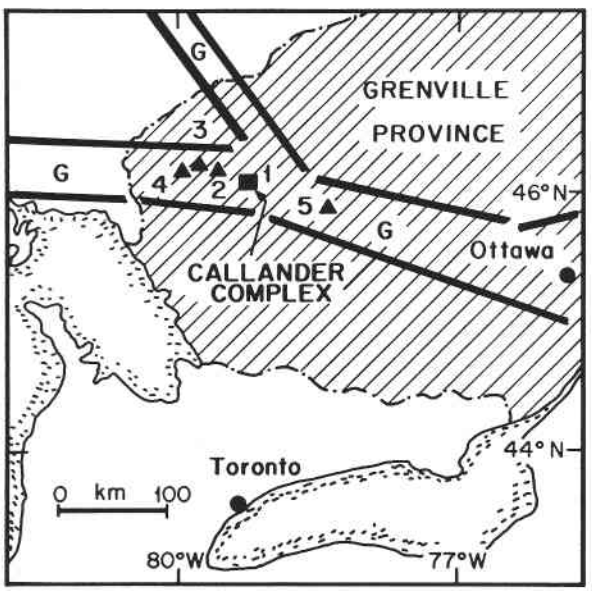


FIG. 1. Location map showing the Ottawa–Bonnechere graben (G) system with the (1) Callander, (2) Manitou Islands, (3) Burritt Island, (4) Iron Island, and (5) Brent Crater complexes.

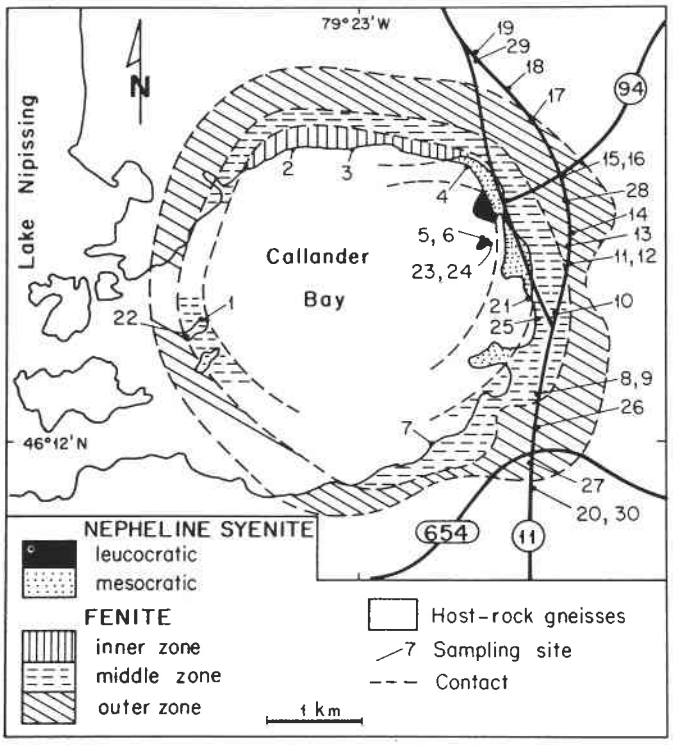


FIG. 2. Geology of the Callander Complex, from Ferguson and Currie (1972), with sampling sites.

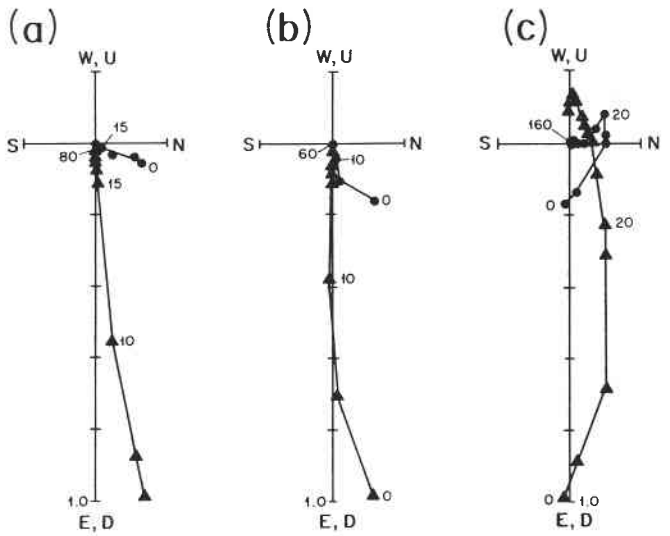


FIG. 3. Orthogonal plot of the AF demagnetization vector for example A-component specimens from sites (*a*) 16, (*b*) 17, and (*c*) 6. Points on the horizontal (N–E–S–W) and vertical (N–D–S–U) planes are denoted by circles and triangles, respectively. The axial intensities are expressed as a ratio of the NRM intensity. The AF intensities are in mT.

in a fine-grained dark green matrix with titanomagnetite (Ferguson and Currie 1971).

Using current decay constants (Steiger and Jager 1977), M. Shafiqullah obtained two identical K–Ar biotite ages of 564 ± 14 Ma for the lamprophyre dikes (*in* Ferguson and Currie 1972). Currie (1976) gave K–Ar biotite ages of 568 Ma for a lamprophyre dike and 575 Ma for the nepheline syenite. Recently, Kamo *et al.* (1989*b*) reported a Pb–Pb isochron age using sphene, brookite, anatase, apatite, and rutile separates of 550 ± 4 Ma for a lamprophyre dike in an abstract, but they increased it to 576 ± 3 Ma on presentation. Agreement between the K–Ar and Pb–Pb ages implies cooling of the pluton within a very few million years with no later metamorphism. The adjacent Manitou Islands complex has given K–Ar ages of 566 ± 38 and 576 ± 45 Ma (Lowdon *et al.* 1963; Gittins *et al.* 1967) and a Brent Crater dike of 582 ± 40 Ma (Shafiqullah *et al.* 1968). Taken together, an age of 575 ± 5 Ma appears reasonable for the Callander Complex.

Currie and Ferguson (1971) estimated that the complex has undergone simple uplift of about 8–12 km since passive emplacement. Analysis of the aeromagnetic anomaly indicates that the pluton is vertical or untilted (West 1989). This agrees with the presence of flat-lying Middle Ordovician limestones unconformably overlying the Manitou Islands complex about 12 km west-northwest of Callander Bay (Lumbers 1971).

Experimental methods

The Callander Complex was sampled by drilling at 19 lamprophyre dike sites and 10 sites in other rock units (Fig. 2; Table 1). One or two specimens were sliced from each of the four or five cores drilled at each site to give a total collection of 252 specimens. The cores were oriented by sun compass.

The natural remanent magnetization (NRM) of each specimen was measured on an automated CTF DRM-420 cryogenic magnetometer inside a shielded room. Two specimens from each site were alternating-field (AF) demagnetized in 13 steps to 160 mT using a Sapphire Instrument SI-4 unit, and a further two specimens were thermally demagnetized in 13 steps to 680°C using a Schonstedt TSD-1 unit. All 15 specimens from site 29 in the host rock were similarly step demagnetized for a contact test. The remaining specimens from each site were AF demagnetized in four selected steps, and those of four sites were thermally demagnetized at 570°C in addition. Stable characteristic directions were accepted if their maximum angular deviation (MAD) was $\leq 10°$; however, the MAD was $\leq 3°$ for most specimens (Kirschvink 1980). These were used, along with remagnetization circle directions in a few sites (Bailey and Halls 1984), to calculate the site mean directions (Table 1). Eight specimens were subjected to isothermal remanent magnetization (IRM) tests by magnetizing them in eight direct-current (DC) field steps to 200 mT and then AF demagnetizing them in six steps to 40 mT.

Results

Natural remanent magnetization

The NRM intensities are typically strong (10^{-2} to $10° A \cdot m^{-1}$) for the lamprophyre sites and moderate (10^{-4} to $10^{-2} A \cdot m^{-1}$) for the other sites. Most NRM directions are normal and steeply inclined about the present Earth's magnetic field (PEMF) direction; however, reversed directions are found at five sites (5,6,10, 12, and 29).

Step demagnetization

Most normal specimens show the initial removal of a Holocene viscous remanent magnetization (VRM) in about the PEMF direction up to about 10 mT on AF step demagnetization and then the emergence of a near-vertical to steeply eastwards normal component in the higher coercivity domains that decays linearly to the origin (Figs. 3*a*,3*b*). The latter is labelled the A_n component. The reversed specimens also show the initial removal of Holocene VRM but up to higher coercivities, and then the isolation of a near-vertical to steeply northwesterly

TABLE 1. Site mean remanence directions

Site	Rk	N_m,	N_e,	N_r	Demagnetization	D (°)	I (°)	α_{95} (%)	k
1	L	9,	8,	0	20–160/450–580	106	74	13	18
2	FI	9,	8,	0	10–60/525–580	89	69	12	24
3*	C	6,	5,	0	0–160/525–630	65	55	25	10
4	SF	12,	12,	0	15–60/300–580	6	85	10	21
5	L	9,	4,	3	20–60/525–580	332	−78	8	54
6	SM	9,	6,	3	60–120/500–580	335	−82	13	17
7	T	6,	5,	0	10–160/600–680	68	75	14	32
8	L	9,	8,	0	10–60/100–580	119	83	12	21
9	L	9,	9,	0	10–80/200–580	5	81	10	30
10	L	11,	8,	3	10–30/300–580	28	−88	9	29
11	L	8,	6,	0	10–60/300–580	36	83	13	29
12	L	7,	7,	0	0–160/200–525	137	87	14	18
13	L	8,	8,	0	20–50/450–580	84	74	8	46
14	L	8,	6,	0	10–30/100–580	125	78	6	117
15	L	10,	9,	0	10–80/500–580	73	79	6	86
16	L	10,	9,	0	10–60/400–580	58	85	4	136
17	L	8,	8,	0	10–60/525–580	95	80	10	31
18	L	8,	8,	0	10–50/300–580	89	83	8	44
19	L	12,	12,	0	10–30/400–580	350	83	5	86
20	L	12,	11,	1	10–30/200–570	116	65	7	36
21	L	10,	9,	0	10–60/400–580	39	88	12	20
22*	FM	4,	1,	2	15–25/400–500	328	−73	9	191
23	SM	6,	5,	0	10–30/450–580	49	76	7	119
24	T	6,	4,	1	15–40/525–580	77	83	9	79
25	FM	8,	6,	0	10–120/525–600	112	87	15	20
26	FO	8,	6,	0	20–120/500–600	58	82	10	44
27	FO	7,	6,	0	20–60/450–570	65	77	12	34
28	FO	6,	6,	0	10–50/450–580	350	77	14	25
29	A	4,	4,	0	5–15/400–500	159	72	31	10
	B	3,	3,	0	10–160/400–660	304	−71	14	81
	C	5,	5,	0	15–80/450–600	293	−67	8	89
	D	3,	3,	0	50–140/0–660	309	−71	14	83

NOTES: The rock types (Rk) are mesocratic and leucocratic syenite (SM, SL); lamprophyre, carbonatite, and trachyte dike (L, C, T); inner, middle, and outer zone fenite (FI, FM, FO); and granite gneiss at 0.1, 0.6, 1.6, and 4.1 m (blocks A, B, C, D) from lamprophyre dike contact at site 19. N_m, N_e, and N_r are the number of specimens measured, giving used end points, and giving used remagnetization circles, respectively. The demagnetization treatment range is designated by the AF range in mT, followed by the thermal range in °C. D, declination; I, inclination; α_{95}, radius of 95% confidence; k, precision parameter (Fisher 1953).

*Site not accepted for calculating means as described in text.

reversed component (A_r) at even higher coercivities (Fig. 3*c*). Thermal step demagnetization gives a similar result. In normal specimens a Holocene VRM is removed up to about 400–450°C, and then A_n is removed up to the 585°C Curie temperature of magnetite (Fig. 4*a*, 4*b*), whereas in reversed specimens the normal VRM persists up to about 500°C, and then A_r is removed up to 585°C (Fig. 4*c*). Only the trachyte dike at site 7 shows A_n clearly in hematite with laboratory unblocking temperatures between 600 and 680°C (Fig. 5*a*), although several fenite sites show small hematite components close to the A_n and A_r directions.

All 28 sites from the Callander Complex gave some specimens with A_n or A_r characteristic directions. Excluding one site (3) with a radius of 95% confidence (α_{95}, Fisher 1953) of >15° and one site (22) with less than four reliable specimen magnetization directions, the remaining 26 sites gave accepted site mean directions (Fig. 6*a*; Table 1).

Petrologic comparison

The site mean directions were formed into three groups, namely the syenitic core (SM, SL, and T sites in Table 1), lamprophyre dike (L sites), and fenite aureole (FI, FM, and FO sites) groups with the A_r directions reversed in polarity. Since each group mean falls within the cones of 95% confidence of the other two group means (Table 2), all three groups can be drawn from one population (McFadden and Lowes 1981). This indicates that all three groups have coeval magnetizations and, given the unaltered character of the syenites and lamprophyres, that they retain a primary remanence because it would take a postintrusive metamorphic event of about upper greenschist rank to reset the high-coercivity and unblocking-temperature domains carrying the A magnetization (Pullaiah *et al.* 1975).

Reversals test

The A_n and A_r site mean directions, after reversing the latter, can be drawn from one population with >95% confidence (Table 2, McFadden and Lowes 1981), and thus they are antiparallel. The test is not robust because there are only three A_r means, so a few degrees change in any one of them could change the result. Nevertheless, the test suggests that the two groups can be combined to form one population and that A_n and A_r are likely primary components that have not been either partially or

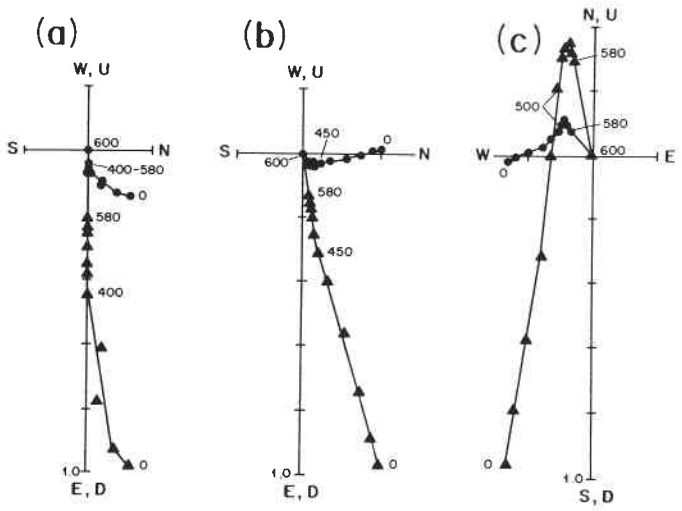


FIG. 4. Orthogonal plot of the thermal demagnetization vector for example A-component specimens from sites (*a*) 14, (*b*) 15, and (*c*) 5. Conventions as in Fig. 3 except the temperatures are in °C.

wholly reset by a secondary metamorphic event because there is no petrologic evidence of regional metamorphism in the Callander Complex.

Contact test

A contact test was run using a 0.7 m wide Cambrian lamprophyre dike (site 19) that intrudes unfenitized Grenville granite gneiss (site 29) (Fig. 2). At 4.1 m from the dike's contact the gneiss retains a moderately steep reversed northwesterly remanence, termed the B component, that resides in hematite with laboratory unblocking temperatures above 620°C (Figs. 5*b*, and 6*b*; Table 1, block 29D). Closer to the dike, the gneiss retains a lower temperature normal component that may be either a Holocene VRM or a partial overprint in low-blocking-temperature domains of A_n from the dike intrusion (Table 1, blocks 29B and 29C). This low-temperature component must be removed before the B magnetization emerges at higher temperatures (Fig. 5*c*). At 0.1 m from the dike's contact in the baked zone, the granite gneiss appears to be remagnetized with the dike's A_n direction in the high blocking temperature domains (Fig. 6*b*; Table 1, block 29A). This positive contact test (Everitt and Clegg 1962) shows that the dike remanence is primary or Early Cambrian because any secondary event would have reset both the dike and host-rock remanence to the same direction. Further, the B-component direction is statistically significantly different from the A_r direction and from the A_n direction after reversal, noting that block means for B are being compared with site means for A (McFadden and Lowes 1981).

IRM analysis

The IRM acquisition curves show behaviours typical of a variety of rock types. Three of the lamprophyre specimens are indicative of multidomain (MD) magnetite (Fig. 7*a*, specimens 9, 15, and 19) and one (5) of pseudosingle-domain (PSD) magnetite. The mesocratic syenite specimen (6) also tracks the PSD type curve. The inner zone fenite (2) and unaltered granite gneiss (29) specimens both show a distributed spectrum from the MD, through PSD, to the single-domain (SD) magnetite fields, with a subsequent contribution from some hematite content. Site 3 in a carbonatite vein did not give a reliable site mean remanence direction, perhaps because the hematite was produced by modern weathering (3). AF demagnetization of the

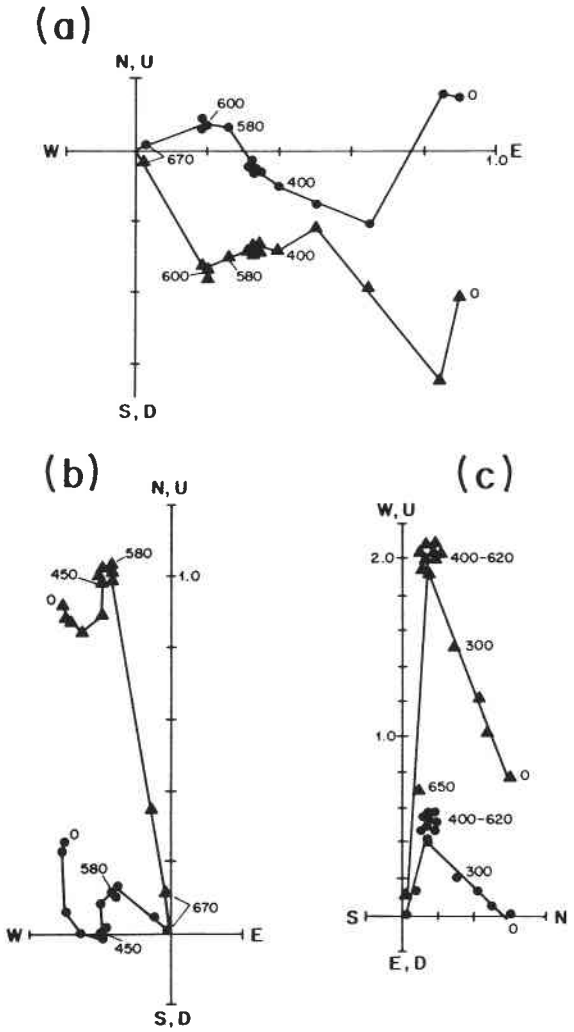


FIG. 5. Orthogonal plot of the thermal demagnetization vector for example specimens from (*a*) site 7 to show the A_n component in hematite, (*b*) site 29D to show the B component in host rock unaffected by dike intrusion, and (*c*) site 29C to show a possible low-unblocking-temperature A_n component in host rock closer to the dike.

IRM also shows the variety of PSD to MD magnetite remanence carriers (Fig. 7*b*). Overall, these IRM results confirm the thermal demagnetization results to indicate that the Callander specimens contain a broad AF coercivity spectrum with suitable fractions of both PSD and SD magnetite and of hematite for carrying a stable primary remanence.

Discussion of the B component

The gneissic host rocks close to the Callander Complex give a Sm–Nd model age of 1590 Ma that indicates a Middle Proterozoic origin for them (Dickin and McNutt 1989). However, the age of metamorphism of the gneisses next to the complex is shown by their location on the 975 Ma isochron derived from K–Ar biotite dating of granitic gneisses in the Grenville Province (Baer 1976; Easton 1986). The pole position for the B component in the granite gneiss at site 29 is located at 22°S, 134°E, which falls directly on the APWP of Dunlop and Stirling (1985) for the Grenville at 980 Ma. The agreement of the radiometric and paleomagnetic ages confirms that neither the host rock nor the complex has been subjected to regional metamorphism since 975 Ma and that the contact test is

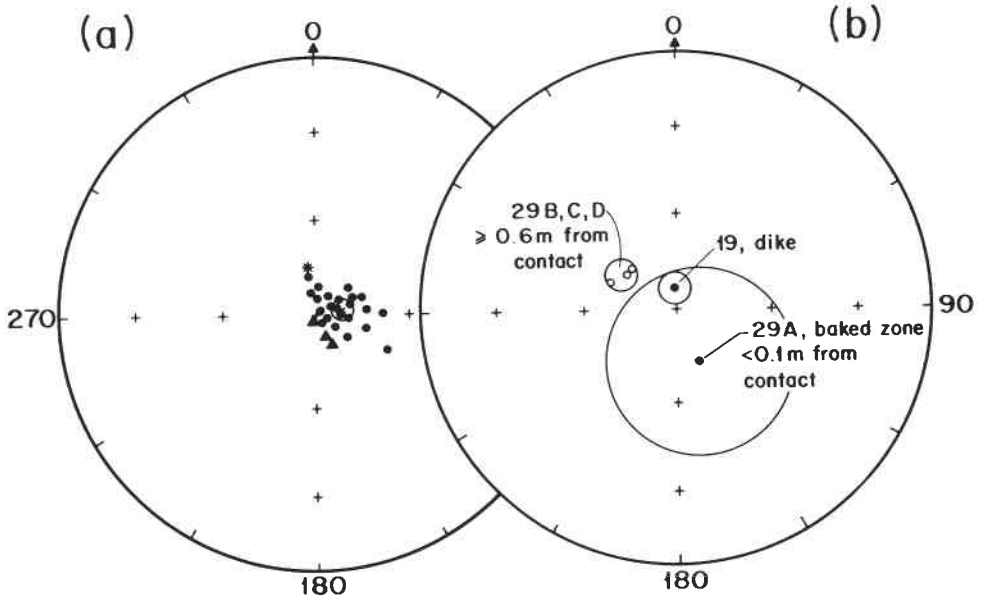


FIG. 6. Equal-area stereogram showing: (*a*) the A_n (●) and A_r (▲ in antiparallel position) site mean directions, with the overall A-component mean (x) circumscribed by its cone of confidence and the asterisk denoting the PEMF direction; and (*b*) the B-component mean from sites 29B, 29C, and 29D, which are remote from the lamprophyre dike (site 19), and from 29A in the dike's baked zone with down (up) directions as closed (open) circles.

TABLE 2. Group mean remanence directions and poles

Group of site means	*N*	*D* (°)	*I* (°)	α_{95} (°)	*k*
Syenites (SM, SL, T)	5	68.2	82.4	7.8	97
Lamprophyres (L)	16	95.1	82.9	4.0	86
Fenites (FI, FM, FO)	5	61.3	80.8	9.9	61
Normal A_n (all but 5, 6, 10)	23	75.8	81.9	3.3	88
Reverse A_r (5, 6, 10)	3	337.6	−82.9	8.5	212
All A	26	82.2	82.7	3.1	83
Host-rock B (29B, 29C, 29D)	3	301.4	−69.8	5.6	478

NOTES: Conventions as in Table 1. A_n, A_r, A, and B are the normal A, reversed A, A, and B magnetization components, respectively. Using a latitude and longitude of 46.2°N and 79.4°W for the complex, the pole position for the A component is 46.3°S, 121.4°E (dp = 5.9°, dm = 6.1°), and for B is 21.6°S, 133.6°E (dp = 8.2°, dm = 9.6°).

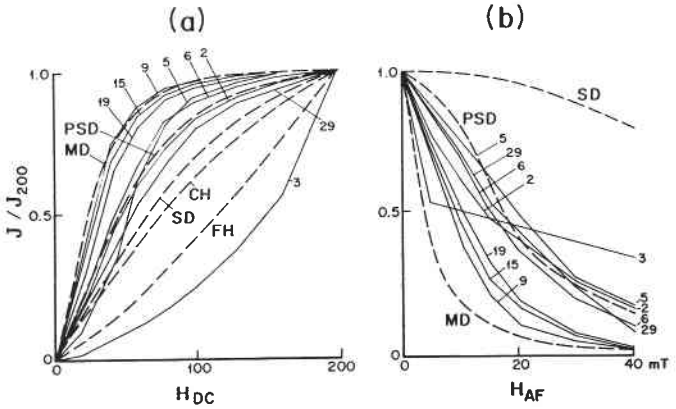


FIG. 7. Isothermal remanent magnetization (IRM) test specimen results showing (*a*) acquisition of IRM and (*b*) AF demagnetization of IRM. The type curves are for single-domain (SD), pseudosingle-domain (PSD), and multidomain (MD) magnetite and for fine-grained (FH) and coarse-grained (CH) hematite (Dunlop 1971, 1972, 1973, 1981).

completely executed or positive because the host-rock remanence predates dike intrusion.

The Cambrian APWP

Definition of a Cambrian (575–505 Ma) APWP for the North American craton has proved elusive. One problem is that the time scale remains unsettled. For example, there are four suggested values, 550, 570, 575, and 584 Ma, for the Precambrian–Cambrian boundary (Smith *et al.* 1989). Another problem is that there are very few latest late Precambrian (625–575 Ma) poles determined for the craton to constrain the older end of the Cambrian APWP (Seguin and Giasson 1987). Yet another is that there are about 30 Cambrian poles that form a broad swath from the southwestern Pacific Ocean to Siberia (Fig. 8). This has led to proposals for an extended northwestwards then southeastwards looping of the Cambrian APWP with very high rates of apparent polar wander that are unlike those of post-Cambrian time (Watts *et al.* 1980; Dankers and Lapointe 1981; Johnson and Van der Voo 1985).

Most of the "Cambrian" poles come from sediments that are susceptible to low-temperature remagnetization by intraformational fluids. Remagnetization of a Cambrian unit was first shown by Al-Khafaji and Vincenz (1971) and has since been shown by stability tests in several other units (e.g., Brown and Van der Voo 1982; Gillett 1982; Peck *et al.* 1986). Remagnetization has also been interpreted in other units to explain multiple populations of remanence directions (e.g., Elston and Bressler 1977; Gillett and Van Alstine 1982; Dunn and Elmore 1985). Noting the general superposition of the Cambrian swath on the Ordovician to Triassic APWP, Johnson and Van der Voo (1985) suggested that remagnetization of Cambrian units is a pervasive effect.

Recently, Van der Voo (1990) has reassessed the Cambrian data, eliminating many poles as either remagnetized or of too low quality, to construct a new Cambrian APWP (Fig. 9*a*). As

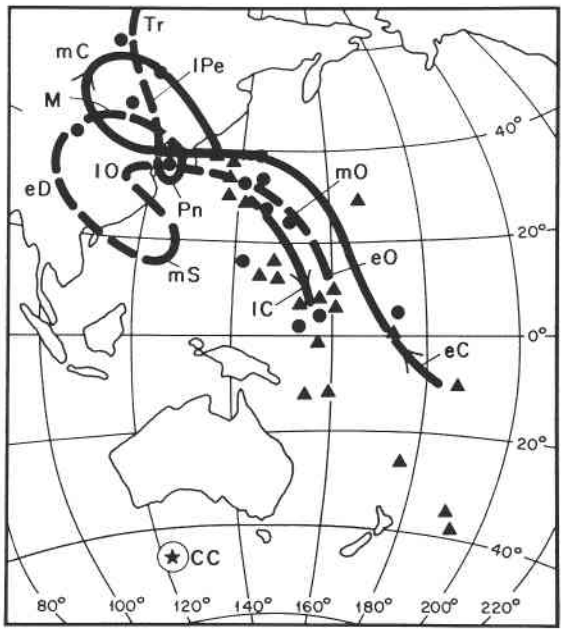


FIG. 8. The swath of pole positions from Cambrian units. Poles used by Watts *et al.* (1980) are shown as circles, and the additional poles used by Dankers and Lapointe (1981) as triangles. One suggested Cambrian APWP (solid heavy line) is shown in comparison with the Ordovician to Triassic APWP (broken heavy line) by Johnson and Van der Voo (1985). ★, Callander Complex pole (CC).

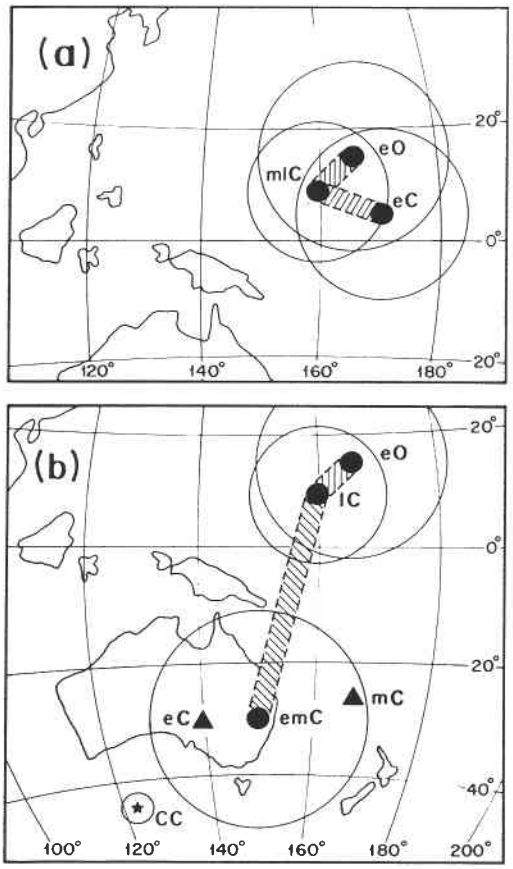


FIG. 9. Cambrian apparent polar wander for North America from (*a*) Van der Voo (1990), showing the Early Ordovician (eO), Middle and Late Cambrian (mlC), and Early Cambrian (eC) positions with their cones of 95% confidence (α_{95}; Fisher 1953); and (*b*) this study, using the following Late Cambrian (lC) poles: Point Peak Member (6°N, 159°E; Van der Voo *et al.* 1976), Morgan Creek Member (11°N, 158°E; Loucks and Elmore 1986), Taum Sauk Member (4°S, 176°E; Dunn and Elmore 1985), McClure Mountain Complex (14°N, 146°E; Lynnes and Van der Voo 1984), and Colorado intrusives (12°N, 156°E; French *et al.* 1977) (average lC is 10°N, 159°E, $\alpha_{95} = 11°$, $N = 5$; ●). The Middle Cambrian (mC) poles include the Bourinot volcanics (21°S, 160°E) and sediments (174°E, 33°S; Johnson and Van der Voo 1985) (average mC is 27°S, 167°E, $\alpha_{95} = 39°$, $N = 2$; ▲). The Early Cambrian poles (eC) include the tilt-corrected Sept-Îles complex B component (44°S, 134°E; Tanczyk *et al.* 1987), Buckingham latites (13°S, 157°E; Dankers and Lapointe 1981), Puerto Blanco volcanoclastics (20°S, 137°E; Barr and Kirschvink 1983), and Callander Complex (46°S, 121°E; this study) (average eC = 31°S, 139°E, $\alpha_{95} = 25°$, $N = 4$; ▲) (average emC is 31°S, 149°E, $\alpha_{95} = 18°$, $N = 6$; ●).

calculated, this path has large error limits. Nevertheless, the Callander pole falls well off this path. This fact leads us to make some cautionary comments about this path and some tenuous speculations about an alternative path that trends southwest towards Australia.

Late Cambrian

The Late Cambrian segment of the APWP must link to the Ordovician segment, which has been relatively well defined by both North American and rotated European poles by Van der Voo (1990) except for the Early Ordovician. We accept the three Late Cambrian poles from the Point Peak Member (Van der Voo *et al.* 1976) and from the underlying Morgan Creek Member (Loucks and Elmore 1986) of the Wilberns Formation in Texas and from the Taum Sauk Member (Dunn and Elmore 1985) of the Bonneterre Formation in Missouri that Van der Voo (1990) accepts. Nevertheless, we note that none of the three are shown to be Late Cambrian by field tests and could be younger Ordovician remagnetizations. In addition, the Morgan Creek pole is based on scattered data, with only three of its nine sites giving α_{95} values of less than 27°.

Van der Voo (1990) accepts as Middle Cambrian one pole from the McClure Mountain syenite complex (Lynnes and Van der Voo 1984) and two poles from its associated dike swarm. French *et al.* (1977) recognized that the case for the two Cambrian poles from one swarm was not strong. We agree and opt to use their combined pole of 12°N, 156°E. Further, the complex has yielded several Late Cambrian Rb–Sr isochron ages, K–Ar mineral ages, and fission-track sphene ages for the syenite but only one Middle Cambrian Rb–Sr isochron age. Thus, we opt to assign both the complex and dike poles to the Late Cambrian. Although there are no contact tests to show that these poles are primary, there is no evidence for later Ordovician orogenic activity to give poles of this ilk. All of these Late Cambrian poles indicate equatorial paleolatitudes for central North America.

Middle Cambrian

The Middle Cambrian Bourinot volcanics and sediments are in the Avalon terrane of Nova Scotia. Their remanence is clearly primary, giving a positive fold test and dual polarities. We speculate that Johnson and Van der Voo's (1985) less favoured option is correct. It suggests that the Avalon terrane was close to the North American craton at this time, with a narrow or nonexistent intervening Iapetus Ocean. Therefore, we accept both poles for the present, subject to later comment.

Early Cambrian

The Sept-Îles Complex A pole of Tanczyk *et al.* (1987) is the

other Middle Cambrian pole accepted by Van der Voo (1990). Higgins and Doig (1981) determined six Rb–Sr isochron ages of 586 ± 24, 579 ± 21, 540 ± 27, 540 ± 7, 535 ± 11, and 475 Ma, mostly from the later acidic phases of this complex. They regard the 475 Ma value as an uplift age. Thus, an Early Cambrian age for the mafic phases that dominate in the paleomagnetic sampling seems more appropriate. Tanczyk *et al.* (1987) regarded the A pole, derived from 10 sites, to be older than the B pole because of a contact test with a pre-Ordovician "late diabase" dike. The A pole, however, provides an excellent fit with the 475 Ma age of the Taconic orogeny. Analysis of gravity data indicates that most of the southern half of this 80 km diameter pluton has been downwarped and overthrust by Taconian nappes (Loncarevic *et al.* 1990), so an overprint from Taconic overthrust burial is likely. Conversely, the Tanczyk *et al.* (1987) B pole was found at 16 sites, including four "late diabase" dikes that Higgins and Doig (1986) have shown geochemically to be comagmatic and coeval with the rest of the complex. Unlike A, the B remanence component is a dual-polarity population, which is an indication of primary remanence. Also, the B pole, which is > 80° away from the A pole, does not fall on any part of the Phanerozoic APWP (Irving and Irving 1982; Gordon *et al* 1984; Van der Voo 1990), as one would expect if it were a remagnetization as Tanczyk *et al.* (1987) proposed. Thus we opt to consider the B remanence, calculated using all 16 sites, to be primary and to correct its pole for a 10°SE tilt that is observed in the contact relations, the petrologic distribution, and the gravity data (Goodacre 1976; Higgins and Doig 1981; Tanczyk *et al.* 1987; Loncarevic *et al.* 1990).

Three of the six Early Cambrian poles accepted by Van der Voo (1990) come from groups I, II, and III of the Buckingham latites. These poles are represented by only two, six, and three paleomagnetic sites, respectively, and possibly by even fewer flows (Dankers and Lapointe 1981). We think this is an unrealistic weighting of three very similar directions and therefore combine all 11 to give one tilt-corrected pole of 13°S, 157°E. This pole, based on a contact test, must be older than Early Ordovician.

Another Early Cambrian pole accepted by Van der Voo (1990) is from a hematite component in the Unicoi basalts. Brown and Van der Voo (1982) attributed the hematite to primary high-temperature oxidation based on the "Cambrian" location at that time of its tilt-corrected pole. However, its uncorrected location is a good fit to the Ordovician segment of the APWP (Van der Voo 1990), and these flows were metamorphosed to greenschist at this time during the onset of the Taconic orogeny. They also retain a postfolding overprint in magnetite. Other Cambrian units, such as the Waynesboro and Rome formations (Watts *et al.* 1980) and three ophiolite complexes in Quebec (Seguin 1982), also appear to carry a Taconic remagnetization.

The Bradore red sandstones in the Humber Zone of western Newfoundland provided the fifth Early Cambrian pole accepted by Van der Voo (1990). Rao and Deutsch (1976) observed specularite and believed that it carried a stable "primary" remanence that was stabilized during diagenesis and lithification. No tests constrain this age, so their remanence could well be a later mid-Ordovician Taconic overprint as well. The sixth pole accepted by Van der Voo (1990) is from the Lower to Middle Cambrian Tapeats sandstone in which Elston and Bressler (1977) found a dual-polarity remanence, favouring a primary origin. However, Gillett (1982) found evidence of protracted diagnetic alteration in this unit that makes a Late Cambrian or even Ordovician age assignment plausible.

Although it is reported in an abstract only, the pole from the Early Cambrian Puerto Blanco volcaniclastics (Barr and Kirschvink 1983) is accepted because of its positive conglomerate test. Finally, we accept the Callander Complex A pole from this study.

Overview

The resulting Cambrian APWP is tenuous (Fig. 9b). Its error limits are still very large, being somewhat smaller for Late Cambrian and somewhat larger for Middle to Early Cambrian than the Van der Voo (1990) path. The latter path represents a very slow rate of apparent polar wander that is fairly comparable to the Cretaceous path for North America (Irving and Irving 1982; Gordon *et al.* 1984), whereas this postulated path suggests a more rapid rate that is fairly comparable to the subsequent Ordovician path (Van der Voo 1990). One virtue of this path is that it moves the North American craton into higher paleolatitudes at the start of Cambrian time, rather than leaving it in equatorial paleolatitudes. This fits much better with the glacial paleoclimatic evidence of widespread late Vendian to earliest Cambrian glaciation (Harland 1983) in Alaska (Allison *et al.* 1981), the Yukon – Northwest Territories (Eisbacher 1981), the Avalon Peninsula of Newfoundland (King 1988), Greenland (Higgins 1981), Spitzbergen (Hambrey *et al.* 1981), Norway (Edwards and Foyn 1981), and Scotland (Spencer 1981) when these terranes were thought to be contiguous (Nance 1987; Piper 1987).

The proposed path includes the Middle Cambrian Bourinot volcanics and sediment poles of the Avalon terrane, assuming the Iapetus Ocean was not yet opened. This is permissible because present radiometric age dates bracket initiation of the opening to between 497 and 615 Ma (Kamo *et al.* 1989*a*). If it was not closed, then the postulated Cambrian APWP is constrained by the Early and Late Cambrian positions only (Fig. 9*b*). Johnson and Van der Voo (1985) argued that the Avalon terrane was separated from North America by a minimum of 1000 km because the two terranes differed in paleolatitude by 20 ± 10°. Using the constrained Early to Late Cambrian path, however, there is no paleolatitude discrepancy, as the paleolatitude decreases from about 70° in Early Cambrian, to about 50° in Middle Cambrian, to about 25° in Late Cambrian, and finally to about 17° in Early Ordovician time. There is, however, an angular separation of the Bourinot mean from the constrained path of about 20° that could be explained by apparent polar wander, by an oceanic separation of about 2000 km, or by a simple anticlockwise angular rotation of Cape Breton Island about a vertical axis by about 25°, which is geologically quite feasible.

Tectonic rotation of terrane blocks may be the major source of scatter in the data set of accepted poles. W. A. Morris (personal communication, 1990) pointed out to the authors that the Buckingham latites, Callander Complex, and Sept-Iles Complex B poles fit the proposed path relatively well. They come from the Precambrian Shield and so have not been subject to block rotation. However, several of the other poles come from terranes that have been subjected to later orogenic deformation and may have undergone block rotations. For example, clockwise rotations of about 15° would swing both the McClure Mountain Complex and Puerto Blanco volcaniclastics poles onto the proposed path. Similarly, anticlockwise rotations of about 25° would swing the Bourinot sediments, Bourinot

volcanics, and Taum Sauk Member poles onto the proposed Cambrian APWP. Thus, between the problems of remagnetization overprints and block rotations, it may be very difficult to establish a rigourous Cambrian APWP. This suggests, in turn, that Cambrian intrusives into the Precambrian Shield are important paleomagnetic targets to define the path.

In conclusion, the definition of the Cambrian APWP remains speculative. This study has contributed to the discussion by suggesting a possible alternative path. Clearly, more work is required to provide a definitive answer.

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